Chapter 3

Printed Circuit Board Production Techniques

Syllabus

Unit-I	**Introduction to Printed circuit board:** Fundamental of electronic components, basic electronic circuits, Basics of printed circuit board designing: Layout planning, general rules and parameters, ground conductor considerations, thermal issues, check and inspection of artwork. **[4 Hours]**
Unit-II	**Electronic design automation (EDA) tools for PCB designing** Brief Introduction of various PCB designing software like EAGLE,KICAD,ORCAD, Selecting the Components Footprints as per design, Making New Footprints, Assigning Footprint to components, Net listing, PCB Layout Designing, Auto routing and manual routing. **[6 Hours]**
Unit-III	**Printed circuit board production techniques** **Photo printing, film master production, reprographic camera, basic process for Single/double sided PCBs photo resists, Screen printing process, plating, relative performance and quality control, Etching machines and material(chemicals) required , Solders alloys, fluxes, soldering techniques, Mechanical operations.** **[5 Hours]**

Printed Circuit Board(PCB) Production Techniques

Photo printing

- In the context of PCB (Printed Circuit Board) designing, "photo printing" typically refers to the process of transferring a circuit pattern onto a photosensitive material, such as a photosensitive film or coated copper board. it involves using a photoresist material that reacts to light.

Here's a simplified overview of the photo printing process in PCB manufacturing:

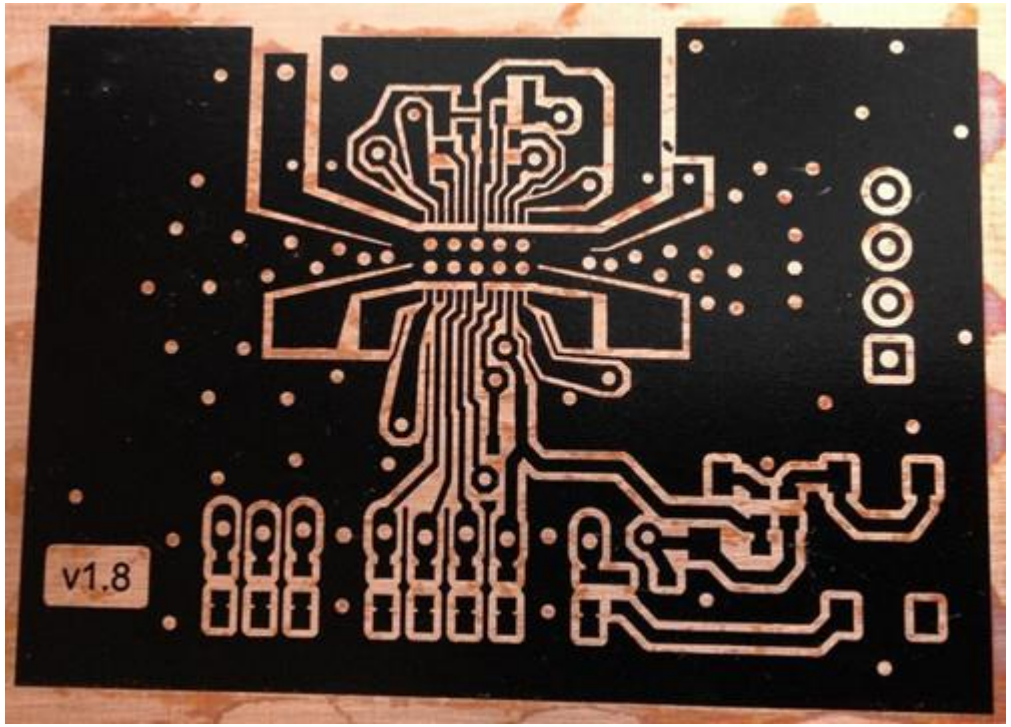
- Designing the PCB Layout: The first step is to create a digital design of the PCB using computer-aided design (CAD) software. This design represents the copper traces, pads, and other elements of the circuit.

- Printing the Photomask: The PCB design is then used to create a photomask. A photomask is essentially a negative of the PCB layout. It is a clear material with opaque areas corresponding to the copper traces and pads on the PCB.

Preparing the Photosensitive Material: The surface of the PCB, usually a copper-clad board, is coated with a layer of photosensitive material, often called photoresist. This material can undergo a chemical change when exposed to light.

Exposure to UV Light: The photomask is placed in direct contact with the coated PCB. The assembly is then exposed to ultraviolet (UV) light. The UV light passes through the clear areas of the photomask and reacts with the photosensitive material on the PCB.

Developing the Image: After exposure, the PCB is developed to remove the unexposed photoresist. This leaves behind the pattern of the circuit on the board.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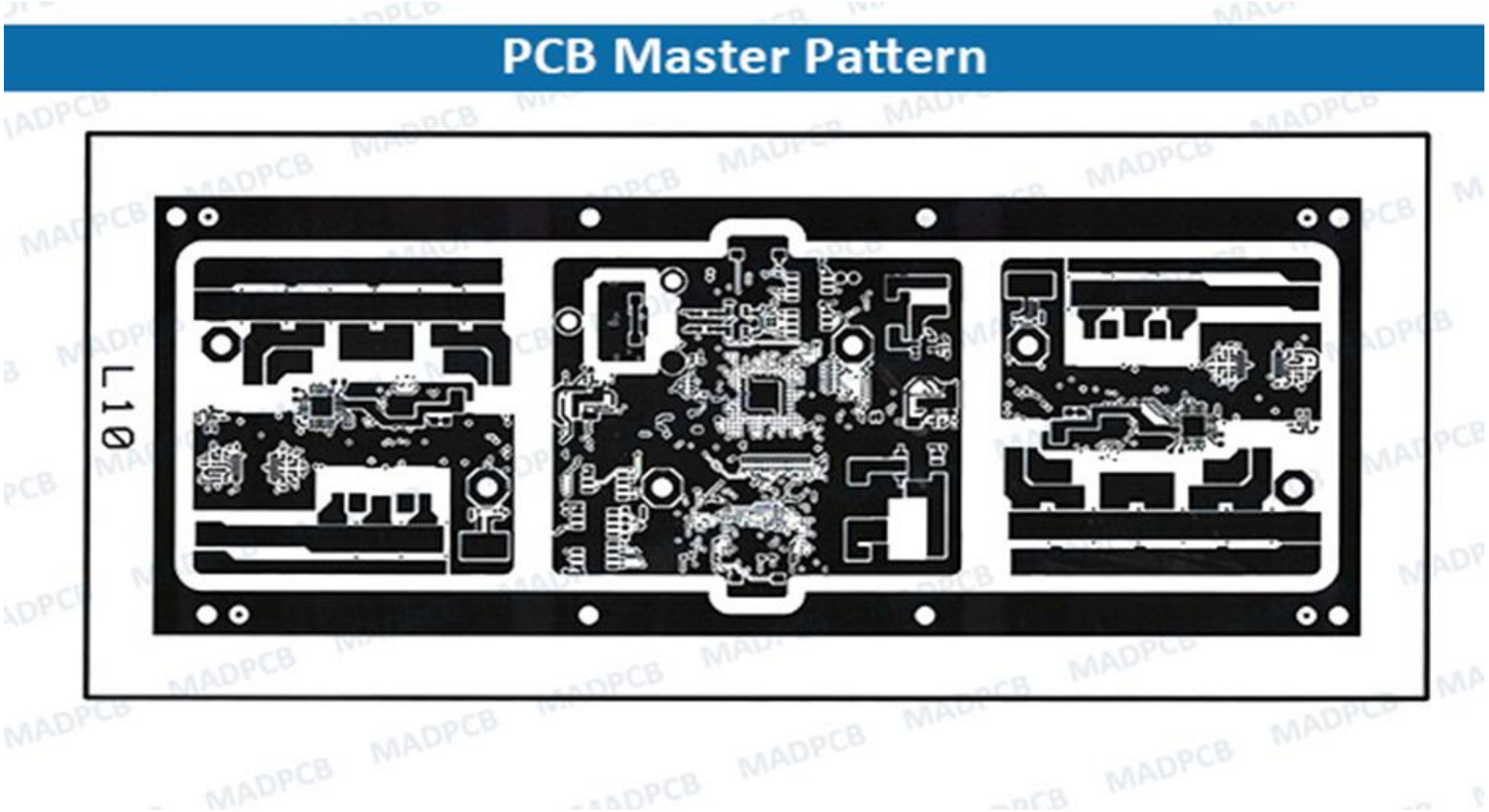
Film Master Production

- In the context of PCB (Printed Circuit Board) design and manufacturing, a "film master" refers to a photographic film that contains the image of the PCB layout. This film master is an integral part of the process used to transfer the circuit pattern onto a light-sensitive material during PCB production.

- The film master (or photoplotter film) is used in the exposure step of the PCB production process. The film master is aligned with the copper-clad PCB substrate coated with photosensitive material, and the assembly is exposed to UV light. The light passes through the transparent areas of the film master, transferring the PCB pattern onto the photosensitive material.

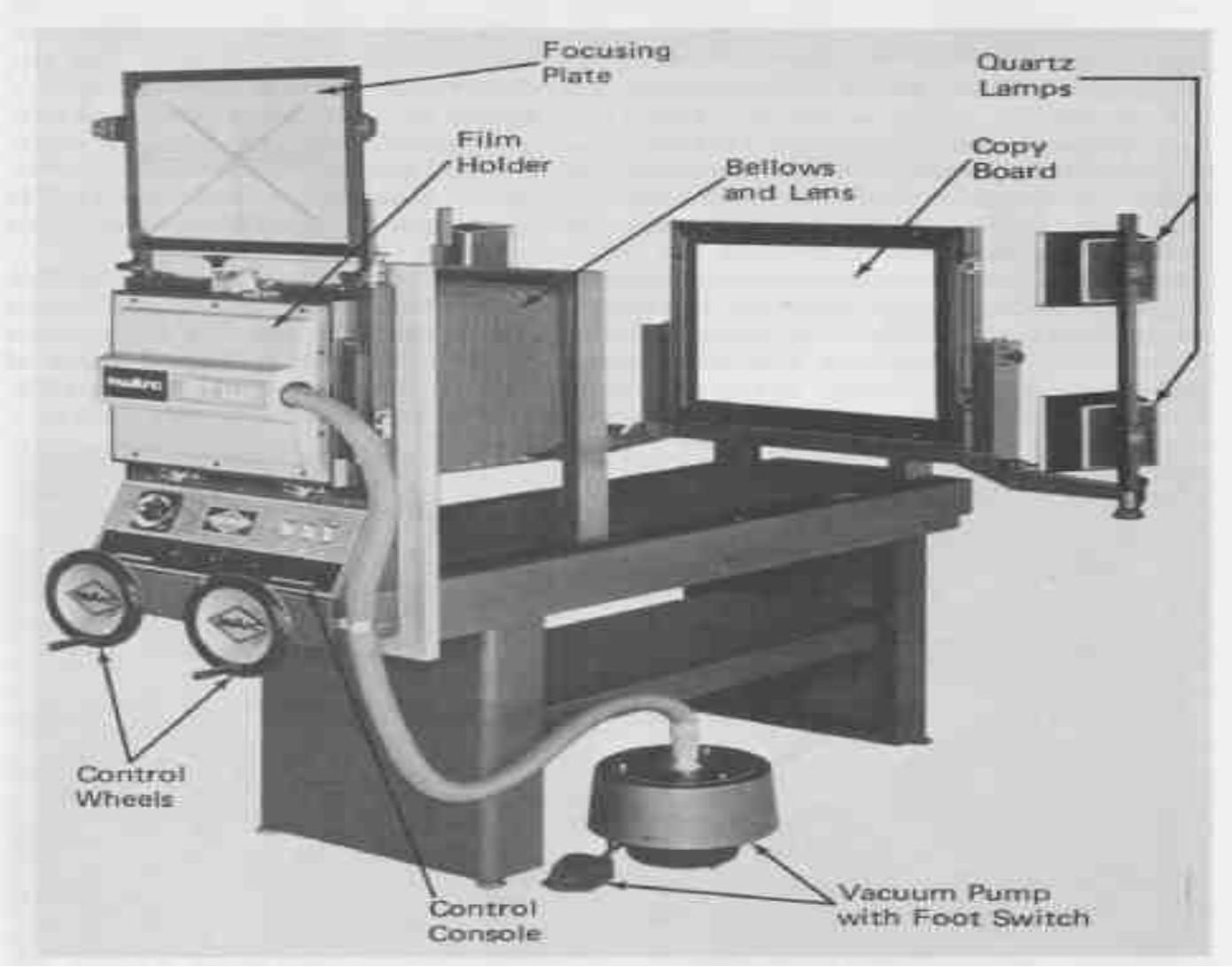
Film Master



Reprographic camera

- A reprographic camera, in the context of PCB (Printed Circuit Board) designing, is a camera used in the process of creating film masters or photoplotters for PCB production. This camera is a crucial tool in the pre-digital era of PCB manufacturing, where the design information was transferred onto photographic film as a negative image.

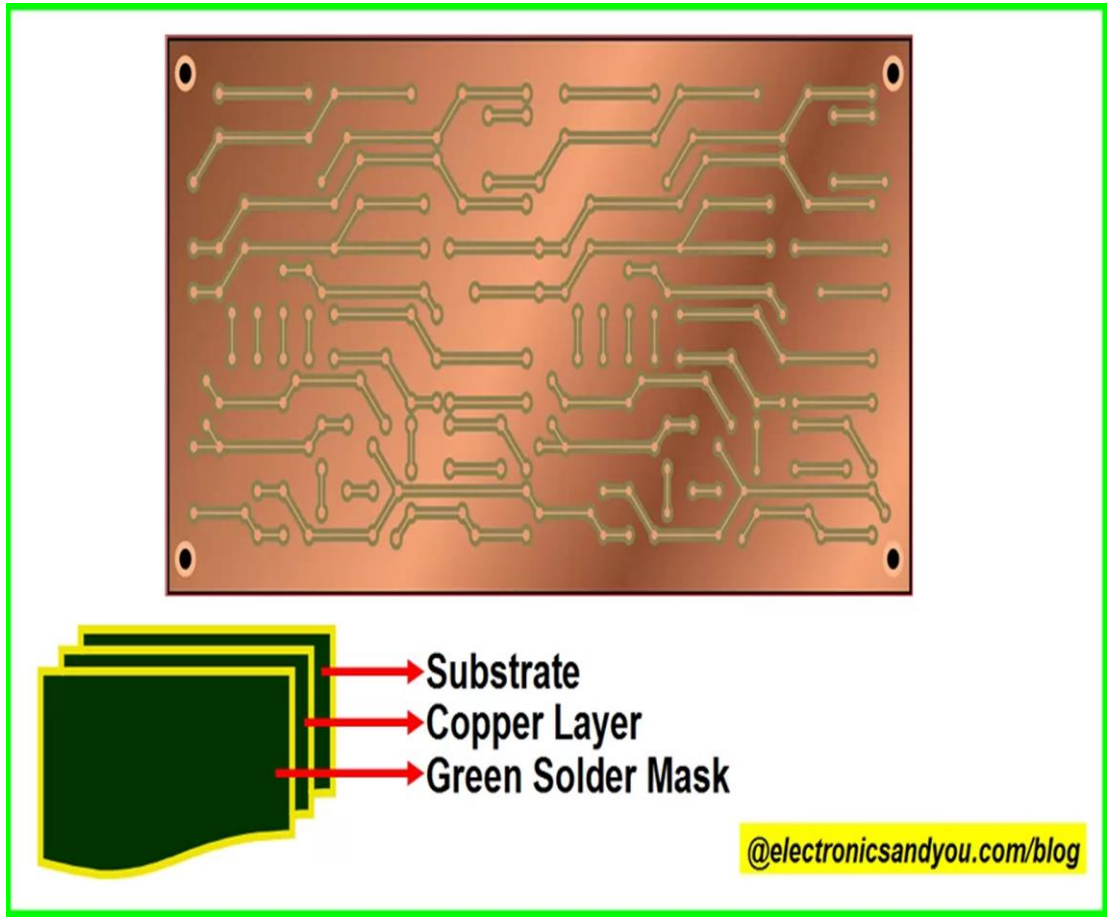
- Film Master Production: The design information is transferred onto a high-contrast photographic film using a reprographic camera. The camera essentially takes a photograph of the PCB layout, creating a negative image on the film. The areas that will become copper traces and pads on the final PCB are transparent on the film, while the areas that will be removed during the etching process are opaque.


Focusing Plate
Quartz Lamps
Film Holder
Bellows and Lens
Copy Board
Control Wheels
Control Console
Vacuum Pump with Foot Switch

Basic process of creating Single /Double sided PCB using photo resist

The process of creating a single or double-sided PCB (Printed Circuit Board) using photoresist involves several steps. This method utilizes a light-sensitive material called photoresist to transfer the PCB pattern onto the board. Here's a basic overview of the process for both single and double-sided PCBs.

Images of Single sided PCB



Process for Developing Single-Sided PCB using Photoresist :

1. PCB Substrate Preparation:

- Start with a copper-clad board, which is typically composed of a layer of copper laminated to a non-conductive substrate (usually fiberglass).
- Clean the copper surface to remove any dirt, oxidation, or contaminants.

2. Applying Photoresist:

- Coat the cleaned copper surface with a layer of photoresist material. This can be done using various methods such as spray coating, dip coating, or roller coating.

3. Pre-baking:

- Heat the board to pre-cure the photoresist. This step removes solvents and prepares the photoresist for exposure.

Single-Sided PCB Photoresist Process:

4. Exposure:

- Place a film mask or a photo tool with the negative image of the PCB layout on top of the photoresist-coated board.
- Expose the assembly to UV light. The light passes through the clear areas of the mask, hardening the photoresist in those regions.

5. Developing:

- Remove the unexposed (softened) photoresist using a developer solution. This reveals the copper areas that will form the traces and pads.

6. Etching:

- Submerge the board in an etchant (usually an acid solution) to dissolve the unprotected copper. The hardened photoresist acts as a mask, protecting the copper traces.

Single-Sided PCB Photoresist Process:

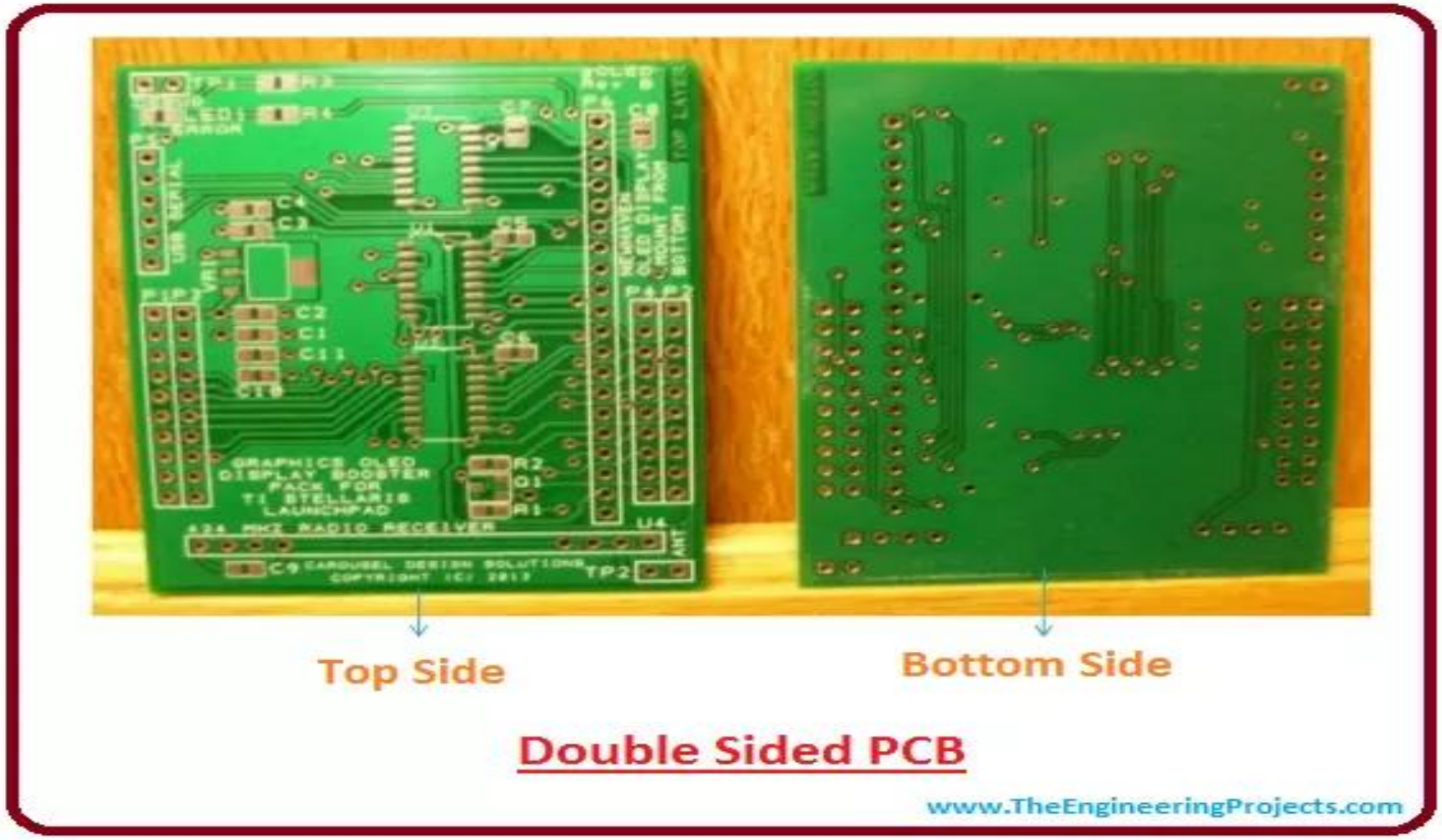
7. Stripping:

- Remove the remaining photoresist, leaving behind the etched copper traces.

8. Cleaning and Finishing:

- Clean the board to remove any remaining chemicals, and inspect for quality. Additional finishing steps may be performed based on the specific requirements.

Images of Double sided PCB



Process for Developing Double-Sided PCB using Photoresist

1. Same as Single-Sided Process:

- Follow steps 1 to 3 from the single-sided process to prepare the PCB substrate, apply photoresist, and pre-bake.

2. Alignment and Exposure (Double-Sided):

- Align a second phototool or film mask with the negative image of the bottom layer on the opposite side of the board.
- Expose the assembly to UV light again, creating the image for the bottom layer.

3. Developing (Double-Sided):

- Develop the exposed board to reveal the photoresist patterns for both the top and bottom layers.

Double-Sided PCB Photoresist Process:

4.Etching and Stripping (Double-Sided):

- Etch the board to remove copper from both sides based on the developed patterns.
- Strip away the remaining photoresist, leaving behind the double-sided PCB with copper traces on both sides.

5. Cleaning and Finishing:

- Clean the board thoroughly and perform any additional finishing steps as needed.
- This process provides a simplified overview, and variations may exist based on specific materials, equipment, and manufacturing requirements.

Screen Printing Process for PCBs:

- Screen printing is a versatile printing technique used for a variety of applications, including fabric printing, and PCB (Printed Circuit Board) manufacturing. Here, an overview of the screen printing process, particularly as it pertains to PCB fabrication:

1. Preparation of PCB Substrate:

- Start with a clean PCB substrate made of a material such as fiberglass with a copper layer laminated onto it.

2. Screen Preparation:

- Create a stencil, also known as a screen, which corresponds to the PCB layout. This is typically done by applying a light-sensitive emulsion onto a fine mesh screen and then exposing it to UV light through a film with the PCB pattern.

3. Aligning the Screen:

- Place the prepared screen over the PCB substrate, ensuring accurate alignment with the desired areas to be printed.

Screen Printing Process for PCBs

- Applying Ink:
- Apply a layer of conductive ink onto the screen, covering the entire stencil.

5. Squeegee Action:

- Use a squeegee to press the ink through the screen onto the PCB. The squeegee moves across the screen, forcing the ink through the openings in the stencil and onto the substrate.

6. Curing the Ink:

- Cure or dry the applied ink on the PCB substrate. This can be done through heat, UV exposure, or other curing methods, depending on the type of ink used.

Screen Printing Process for PCBs

- Repeat for Multiple Layers:
- For multilayer PCBs, the process is repeated for each layer. Different screens are used for each layer, and the alignment is crucial to ensure proper layer-to-layer registration.
- 8. Finalizing the PCB:
- After all layers are printed and cured, the PCB may go through additional processes such as etching to remove excess copper, drilling for component holes, and adding surface finishes.

PCB Etching

- What is PCB etching?

PCB etching is the process of removing unwanted copper from a printed circuit board. Once all of the excess copper has been removed from the PCB, only the required circuit remains. Before the etching process begins, a layout for the board is generated.

- **Types of PCB Etchants**
- Ammonium persulfate ($(NH_4)_2S_2O_8$)
- Cupric chloride ($CuCl_2$)
- Ferric chloride ($FeCl_3$)
- Hydrochloric acid (HCl) mixed with hydrogen peroxide (H_2O_2)
- Sodium persulfate ($Na_2S_2O_8$)

Etching process

- The PCB design is exposed on the copper-clad board. The board is then submerged with an etchant, a chemical solution that selectively removes the unprotected copper. The most common etchants are ferric chloride ($FeCl_3$) or ammonium persulfate ($(NH_4)_2S_2O_8$).

- The etching process continues until all the unwanted copper is dissolved, leaving behind only the copper traces that correspond to the circuit design. The etching time and temperature are critical factors that must be carefully controlled to ensure accuracy and precision.

PCB (Printed Circuit Board) Etching machine

- A PCB etching machine is a equipment designed to remove unwanted copper from a printed circuit board (PCB) during the manufacturing process. The etching machine plays a crucial role in creating the copper traces that form the conductive paths on the PCB.

Types of Etching Machines:

- Manual Etching Machines: These are suitable for small-scale or prototyping purposes. They require more hands-on involvement, as the operator is responsible for applying the etchant and monitoring the process.

- Semi-Automatic Etching Machines: These machines automate some parts of the etching process, making them more efficient than manual machines. Operators may still need to handle certain tasks, such as loading and unloading the PCBs.

- Automatic Etching Machines: These machines are fully automated and can handle the entire etching process without much manual intervention. They are suitable for high-volume production, providing efficiency and consistency.

Solder

- Definition: Solder is a metal alloy with a low melting point that is used to join two or more metal surfaces together. In PCB manufacturing, solder is commonly used to create electrical connections between electronic components and the conductive traces on the board.

- Composition: The most common type of solder used in PCB assembly is a tin-lead alloy. However, due to environmental concerns, lead-free solder alloys have become more prevalent. Common lead-free solder alloys include tin-silver-copper (Sn-Ag-Cu) and tin-copper (Sn-Cu).

- Melting Points: The melting point of solder is a critical factor. It must be low enough to melt without damaging electronic components during soldering but high enough to ensure a stable joint afterward.

Solder Alloys

- Lead-Tin Solder Alloys: Traditional solder alloys contain lead, which has excellent soldering properties. Common lead-tin solder alloys include 63% tin and 37% lead (Sn63Pb37). These alloys have a low melting point and provide good wetting characteristics.

- Lead-Free Solder Alloys: To address environmental concerns and comply with regulations such as RoHS (Restriction of Hazardous Substances), lead-free solder alloys are used. Common lead-free alloys include Sn-Ag-Cu (tin-silver-copper) and Sn-Cu (tin-copper).

Flux

- Definition: Flux is a chemical substance used in soldering to promote the wetting of the solder and enhance the soldering process. It helps remove oxides from the metal surfaces, improves solder flow, and prevents the formation of new oxides during soldering.

Types of Flux

- Rosin Flux: Derived from pine tree resin, rosin flux is a common type used in electronics. It is available in different activity levels (from mildly activated to highly activated) and can be either rosin-core solder or a separate flux applied before soldering.

- Water-Soluble Flux: This type of flux is water-soluble and is often used in applications where post-solder cleaning is required. It can be easily washed away with water after soldering.

- No-Clean Flux: No-clean flux leaves minimal residue after soldering, and in many cases, the residue is non-conductive. It is often used in applications where cleaning is not practical or necessary.

Soldering techniques

Soldering is a fundamental technique in electronics for joining components to printed circuit boards (PCBs) or connecting wires. There are various soldering techniques, each suited to specific applications. Here are some common soldering techniques:

- Through-Hole Soldering:
- Surface Mount Soldering:
- Hand Soldering:
- Wave Soldering:
- Reflow Soldering:
- Drag Soldering (Drag Technique):
- Hot Air Soldering (Hot Air Rework):

Through-Hole Soldering:

Description: Through-hole components have leads that pass through holes in the PCB, and solder is applied on the opposite side of the board.

Technique:

- Insert the component leads through the holes.
- Bend the leads slightly to hold the component in place.
- Flip the PCB and solder from the opposite side, ensuring good contact between the lead, pad, and solder.

Surface Mount Soldering

Description: Surface mount components are mounted directly onto the surface of the PCB, without leads passing through holes.

Technique:

- Apply solder paste to the pads using a stencil.
- Place the components on the solder paste.
- Heat the entire PCB using a reflow soldering oven, which melts the solder paste and forms the solder joints.

Hand Soldering

Description: Hand soldering is a manual process where a soldering iron is used to melt solder and create electrical connections.

Technique:

- Heat both the component lead and the pad on the PCB with the soldering iron.
- Apply solder to the joint, allowing it to flow and create a secure connection.
- Remove the soldering iron and let the joint cool.

Wave Soldering:

Description: Wave soldering is a mass soldering technique primarily used for through-hole components.

Technique:

- Components are inserted into the PCB.
- The PCB is passed over a wave of molten solder, which makes contact with the exposed leads on the bottom side of the board.

Reflow Soldering

Description: Reflow soldering is commonly used for surface mount components and involves pre-applying solder paste to the PCB.

Technique:

- Apply solder paste to the PCB pads using a stencil.
- Place surface mount components on the solder paste.
- Use a reflow soldering oven to heat the entire assembly, melting the solder and forming the connections.

Drag Soldering (Drag Technique)

Description: Drag soldering is a technique used to solder multiple leads of fine-pitch components simultaneously.

Technique:

- Apply a small amount of solder to the tip of the soldering iron.
- Drag the soldering iron across the leads of the component, allowing the solder to flow and form connections.

Hot Air Soldering (Hot Air Rework)

Description: Hot air soldering is often used for SMD (Surface Mount Device) rework.

Technique:

- Apply flux to the area.
- Heat the solder joints with a hot air rework station until the solder reflows and the component can be repositioned or removed.